

Multiple Choice (10 marks)

- Helium effuses through a pinhole 5.33 times faster than an unknown gas. That gas is most likely
 - Ne
 - CO₂
 - Ar
 - CH₄
 - O₂
- Which of the following ions possess a rare gas electronic configuration? (a) Ca^{2+} ; (b) In^{+} ; (c) Ga^{3+} ; (d) Sc^{3+} ; (e) Se^{2-}
 - In^{+} , Sc^{3+} and Se^{2-}
 - Ca^{2+} , Ga^{3+} and In^{3+}
 - Ca^{2+} , Sc^{3+} and Se^{2-}
 - In^{3+} , Ga^{3+} and Se^{2-}
 - Ca^{2+} , In^{+} and Se^{2-}
- Which of the following substances has an asymmetrical molecular structure?
 - CO
 - CO₂
 - CH₄
 - BF₃
 - PCl₅
- Calculate the Coulombic barrier in the bombardment of ^{209}Bi
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 - 22.1 MeV
 - 23.3 MeV
 - 20.5 MeV
 - 21.2 MeV

The atomic radius of hydrogen is 0.037 nm, Compare this figure with the length of the first Bohr radius. Explain any differences.

- The atomic radius of hydrogen is five times the first Bohr radius

- (b) The atomic radius of hydrogen is four times the first Bohr radius
- (c) The atomic radius of hydrogen is half the first Bohr radius
- (d) The atomic radius of hydrogen is one tenth the first Bohr radius
- (e) The atomic radius of hydrogen is larger than the first Bohr radius

True / False (10 marks)

Answer True or False

- 6. True or False: Decomposing 454 g of water by electrolysis requires 125.6 faradays of electricity.
- 7. True or False: Mass spectrometry data supports the existence of isotopes, providing insights into atomic mass variations among elements.
- 8. True or False: $(PV) / (RT) = 1$ suggests a derived relationship that does not correctly represent the ideal gas equation.
- 9. True or False: The ionic bond strength of KF is stronger than that of NaCl due to the larger ionic radius of K^+ compared to Na^+ .
- 9. True or False: The dipole moment of HCOOH, with an OCOH dihedral angle of 0° , is calculated to be 1.41 D.

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the concept of half-life in radioactive decay. Given that 75% of an unstable substance decomposes in one hour, calculate its half-life and explain your reasoning.
- 12. Discuss the relationship between pressure, density, and temperature in interplanetary space, and calculate the temperature necessary for a density of 4 molecules per cubic centimeter at 10^{-16} torr.

End of Paper